

Unit ID: 67adbec96de7f1459b25a5a4

Unit 3 - Alert Cane Creation

PDF Documents

PDF Document 1: 1739439814839-Chapter 5 - unit 3 - Alert Cane Creation.pdf

This PDF document is included in the unit materials.

Update #1 - 2025-05-16 15:31:24



UNIT 3 ALERT CANE CREATION



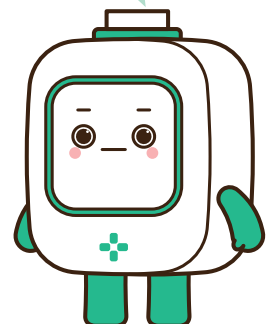
WHAT IS AN ALERT CANE?

An alert cane, a crucial tool designed to aid people with disabilities, particularly those with visual impairments, serves the vital function of signaling for assistance when needed, thereby enhancing their mobility and safety.

We are going to use ToF (Time of Flight) sensors to create an activity that alerts us when there is an obstacle in front.

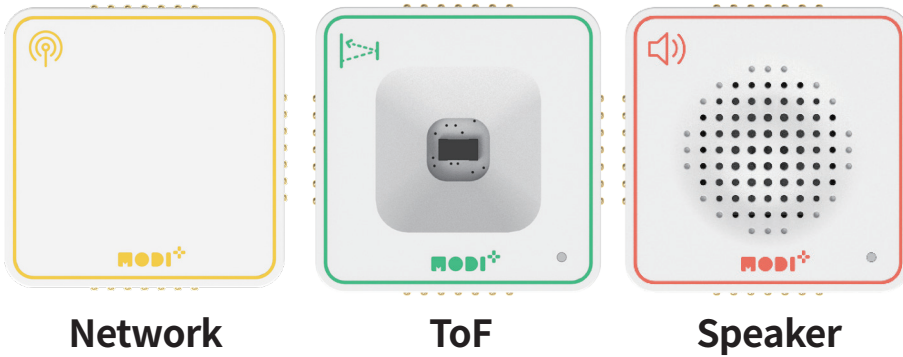
DO YOU REMEMBER WHAT TOF DO?

The role of the ToF (Time of Flight) sensor is to measure distance. You need to move your hand above the sensor to measure distance accurately.



Activity: Create Your Own Alert Cane

1 Materials Needed:



Why is This Important?

Creating an alert cane is a journey of discovery. It's not just about understanding how technology can assist people with disabilities. It's about the joy of learning and the thrill of creating something that can change someone's life. By designing and building this cane, we learn about the importance of accessibility and how we, as a community, can use technology to help those around us. It's a responsibility we all share, and this project is a tangible way to show we care.





Coding

Code Story



Topic Alert Cane



STORY DESIGN

Let's help who can not see
How can we help them?



STORY TO CODE

Which module should be used
to detect obstacles in front?
And how can an alarm be cre-
ated?
Please write down the names
of the modules you can use.



CODE DESIGN

Let's help who can not see
How can we help them?

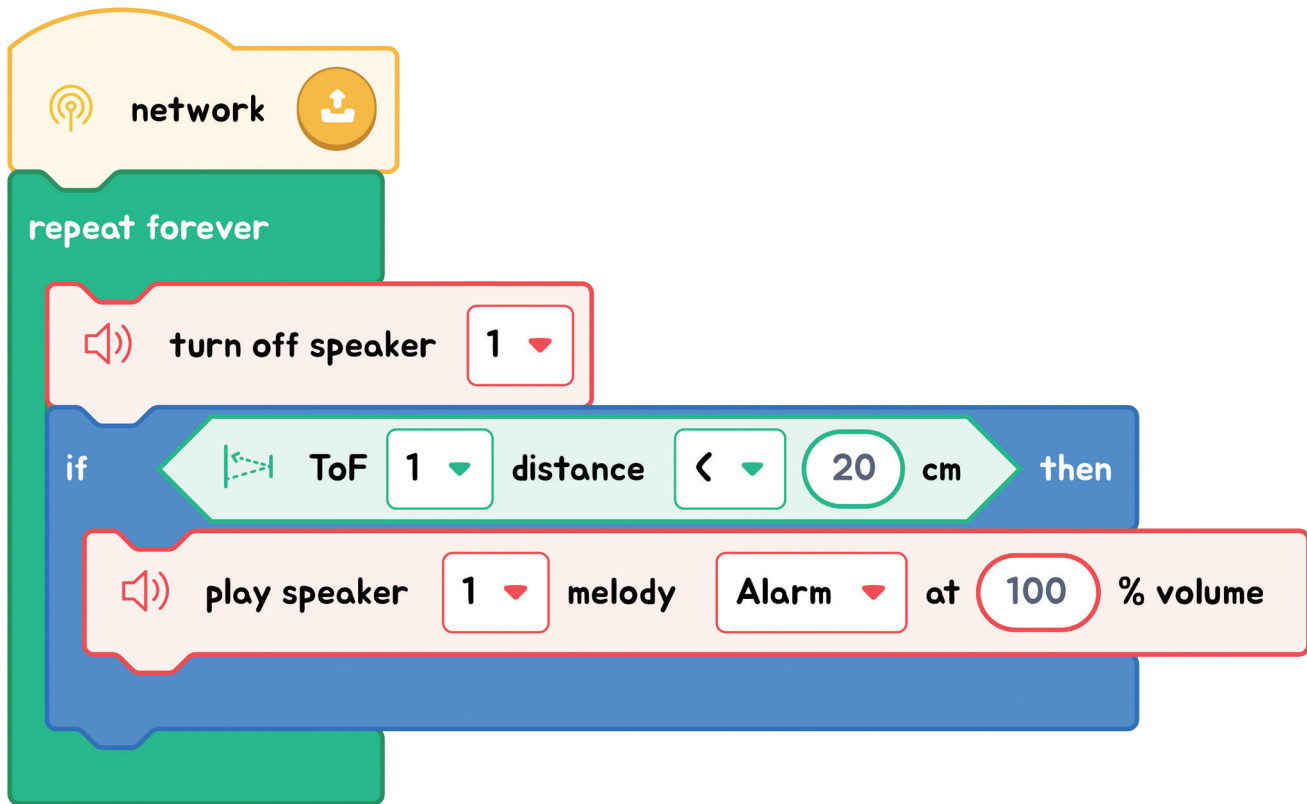


When an obstacle is
detected,



Play an alarm sound.

Code Quest



Test Your Cane:

- Move the cane around and see how the ToF sensor detects obstacles.
- Notice how the speaker sounds when an obstacle is closed, alerting the user.

